Can a Slower Variance Network Stabilize KSIVI?

Learning rates, shared features, and optimization budgets on the X-shaped target

StatComp research investigation

12 October 2026

Abstract

We test variance-specific Adam learning rates and variance architectures in the original StatComp KSIVI implementation on the X-shaped target. The campaign contains 190 primary runs at 50K updates and 36 checkpoint continuations to 200K. A shared variance-head LR of 10^{-5}, with mean/backbone LR 10^{-3}, reduces held-out sliced-Wasserstein error from 0.870 ± 0.524 to 0.129 ± 0.044, improving all five paired seeds. Its mean error is comparable to global variance (0.159); five seeds do not establish superiority. An independent linear variance map also performs well, while a large separate network is less reliable. Exact gradient-path identities, a finite-budget Adam reachability bound, and unchanged-marginal gradient-noise results for constant and Gaussian kernels explain why slower variance updates can help, why excessive slowing can fail, and why head LR does not control the entire shared variance function. Estimator, initialization, annealing, and continuation ablations delimit the result; there is no universal recommendation to decrease the variance LR.

1 Question, findings, and the scope of the evidence

The motivating suggestion is to reduce the variance learning rate while retaining the mean learning rate. That suggestion has two distinct implementations: slowing only a variance output layer on shared features, and slowing an entire independent variance network. They need not have the same effect. We return to the original StatComp model and training policy; the authors' alternative defaults are not substituted for that policy in this campaign.

1. **The proposed slower head helps this conditional setting.** Reducing only the shared variance-head LR by 100× improves all five held-out seeds and reduces their mean error by 85.2%. The backbone retains both mean and variance gradient paths.

2. **The useful rate depends on the variance architecture.** Shared ratio 0.01 and separate ratio 0.03 were selected on discovery seeds. A separate linear variance model selects ratio 1; ratios 0.001–0.1 under-adapt in that model. Separation alone does not establish an advantage over global variance.

3. **Comparable marginal accuracy does not imply the same mechanism.** On held-out seeds the slow original estimator is competitive with global variance and Stein loss. A smaller head LR worsens the tested Stein treatment, so estimator changes and step-size changes are not interchangeable.

4. **Two distinct theoretical effects are necessary.** Smaller block steps reduce update noise at a fixed state, but must still permit the component variance to adapt from its initial value 1.3133. Shared features provide an additional full-rate route, and cumulative LR constrains what a slow independent network can reach.

50K configuration	Discovery (3)	Held out (5)
Shared, equal LR	2.016 ± 1.138	0.870 ± 0.524
Shared; LR ratio 0.01	0.178 ± 0.086	0.129 ± 0.044
Separate 128x2; LR ratio 0.03	0.122 ± 0.062	0.240 ± 0.159
Linear variance; LR ratio 1	0.101 ± 0.032	0.129 ± 0.017
Global variance	0.197 ± 0.130	0.159 ± 0.079
Shared, Stein loss	0.278 ± 0.151	0.262 ± 0.178
Shared, floor 0.2	0.378 ± 0.020	0.397 ± 0.012
Separate, equal LR	0.915 ± 0.516	0.589 ± 0.251

The statistical unit is an independently trained model. Monte Carlo samples from one checkpoint do not count as additional training replications. Discovery seeds select hyperparameters; held-out seeds assess those choices. Descriptive means and sample standard deviations are reported, together with paired per-seed outcomes. This is an exploratory investigation, not a preregistered confirmatory significance test. No statement below establishes a universal ordering of optimizers, architectures, or gradient estimators.

2 The common model and loss

Let $Z \sim N(0, I_2)$ and $U \sim N(0, I_2)$ be independent. StatComp samples

$$X = \mu_\phi(Z) + \sqrt{v_\phi(Z)} \odot U, \qquad v_\phi(Z) = \max\{10^{-4}, \mathrm{softplus}(a_\phi(Z))\}.$$

The mean uses two width-128 SiLU hidden layers. The global model replaces $v_\phi(Z)$ by two learnable variance scalars. At initialization all matched comparisons have the same mean function and constant variance $\mathrm{softplus}(1) = 1.3132616$, except explicitly labeled initialization ablations. The effective mixing dimension is two: the production target binding overrides the raw target-configuration entry.

Write $s_p = \nabla \log p$ and $s_c(X, Z) = \nabla_X \log q_\phi(X \mid Z)$. The conditional Gaussian layer returns its negative score $-s_c = U/\sqrt{v}$. With two independent batches of size $N = 128$, the original loss estimator is

$$\widehat{L}_c = \frac{1}{N^2} \sum_{i,j} k_h(X_i, Y_j) \left(\alpha_t s_p(X_i) + \frac{U_i}{\sqrt{v(Z_i)}} \right)^T \left(\alpha_t s_p(Y_j) + \frac{U'_j}{\sqrt{v(Z'_j)}} \right). \tag{1}$$

For a kernel fixed independently of these samples, the Fisher identity $\mathbb{E}[s_c \mid X] = s_q(X)$ shows that its expectation is the squared KSD between q_ϕ and $p_{\alpha_t} \propto p^{\alpha_t}$, under the usual integration and differentiation conditions. This is the tractable conditional-score formulation underlying KSIVI [1]. A larger conditional family need not have a larger minimum population loss; our question concerns finite stochastic optimization.

For the Gaussian kernel $k_h(x, y) = \exp[-\|x - y\|^2/(2h^2)]$, integration by parts gives the conventional Stein form

$$\widehat{L}_S = \frac{1}{N^2} \sum_{i,j} k_h(X_i, Y_j) \left[s_i^T s'_j + \frac{(s_i - s'_j)^T (X_i - Y_j)}{h^2} + \frac{2}{h^2} - \frac{\|X_i - Y_j\|^2}{h^4} \right], \tag{2}$$

where $s_i = \alpha_t s_p(X_i)$. For independently fixed h, the two forms have the same population objective, but different finite-batch gradients. Neither form has a universally smaller gradient covariance.

Production training fits $h^2 = \text{median}_{ij} \|X_i - Y_j\|^2/[2\log(N+1)]$ on each batch. Original-loss runs differentiate this fitted width. Stein runs hold it fixed during backpropagation, matching the earlier investigation. Thus the default Stein comparison changes both estimator and bandwidth differentiation. We include original-loss bandwidth-detachment controls and independently fixed bandwidth probes; the fixed-kernel population identity is not asserted for a sample-adaptive differentiated statistic.

2.1 Relation to the original KSIVI guarantees

The KSIVI analysis assumes controlled model derivatives and a uniform positive lower bound on component standard deviations (Assumption 4.3), then establishes gradient-variance bounds and an SGD stationarity guarantee under its full assumptions (Theorems 4.7–4.8) [1]. Our small-variance limit leaves a fixed uniformly nondegenerate parameter domain; it does not contradict those results. The implemented 10^{-4} variance floor truncates that limit but can still permit large noise constants. Stationarity also does not imply a small transport error. Adam, annealing, and fitted bandwidths require further care before applying a fixed-objective SGD theorem to these runs.

There is an additional exact boundary on this particular target: its log-density does not have a globally bounded Hessian. In fact, with $c = 1.8/0.76$,

$$\log p(x,y) = C - \frac{x^2+y^2}{0.76} + \log\cosh(cxy), \qquad \partial_{yy}\log p(R,0) = -\frac{2}{0.76} + c^2R^2.$$

The last expression is unbounded as $R \to \infty$, so the target does not satisfy the global Hessian condition in Assumption 4.2. Spatial autograd of the actual target implementation checks the formula. This is a limitation on a direct theorem application, not an explanation of the ordering of methods: all methods in this campaign share the same target.

3 What a variance learning rate can and cannot control

3.1 A head-only learning rate leaves another path active

For a shared backbone $h_b(Z)$, write

$$\mu(Z) = M_\psi h_b(Z) + m_\psi, \qquad a(Z) = W_\omega h_b(Z) + c_\omega.$$

The backbone gradient decomposes exactly as

$$g_b = J_b\mu^T\partial_\mu\widehat{L} + J_b a^T\partial_a\widehat{L} = g_b^\mu + g_b^a. \tag{3}$$

A smaller learning rate for (W, c) leaves the second term in the backbone's gradient active at the backbone learning rate. Mean-driven backbone updates also change the features read by the variance head.

Proposition 3.1 (Exact functional drift decomposition)**.** *For a fixed probe input, let $h' = h + \Delta h$, $W' = W + \Delta W$, and $c' = c + \Delta c$ denote the values before and after one optimizer step. Then*

$$\Delta a = \underbrace{\Delta W h + \Delta c}_{\textit{head}} + \underbrace{W\Delta h}_{\textit{backbone}} + \underbrace{\Delta W\Delta h}_{\textit{interaction}}. \tag{4}$$

Consequently, freezing a trained variance head does not generally freeze its predicted variance on a changing shared backbone.

Proof. Expand $(W + \Delta W)(h + \Delta h) + (c + \Delta c) - (Wh + c)$. When $\Delta W = \Delta c = 0$, the remaining term is $W\Delta h$, which need not vanish. At our constant initialization $W = 0$, this route is initially absent; after the head has learned, it can be present. □

At a fixed state, a first-order SGD calculation gives $\Delta a \simeq -\eta_v J_\omega a\, g_\omega - \eta_b J_b a\, g_b$. As $\eta_v \to 0$ with η_b fixed, its conditional covariance tends to $\eta_b^2 J_b a \operatorname{Cov}(g_b) J_b a^T$ when this matrix is nonzero. Cross covariances matter at finite η_v; the formula is not a universal positive lower bound for every model or state. It does show why head LR alone cannot guarantee control of functional variance noise.

For an independent variance network $a_\omega(Z)$, all of its parameters receive η_v. To first order, both functional drift and stochastic standard deviation scale with η_v at a fixed state. This is genuine control of the whole variance function, while also slowing its deterministic learning. The gradient-path and trained-head-freeze probes test these routes directly rather than inferring them from the final sample plots.

Our stop-gradient ablation detaches the shared features only along the variance path. It removes g_b^a but does not prevent h_b from moving under g_b^μ. Moreover, this deliberately omits part of the shared model's gradient: it is a routing heuristic, not an unbiased gradient of the original shared objective. Separating backbones retains the full gradient of the new parameterization.

3.2 Different rates do not create gradient-estimation bias

Proposition 3.2 (Block learning rates and descent)**.** *Let L have a β-Lipschitz gradient, and let a stochastic gradient $\widehat{g}$ at a fixed state have mean $g = \nabla L$ and covariance Σ. For a deterministic positive block-diagonal learning-rate matrix D,*

$$\mathbb{E}[D\widehat{g}] = Dg,$$

and

$$\mathbb{E}[L(\theta - D\widehat{g})] \le L(\theta) - g^T Dg + \frac{\beta}{2}\{g^T D^2 g + \operatorname{tr}(D^2\Sigma)\}. \tag{5}$$

Proof. Apply the smoothness inequality to $-D\widehat{g}$, take expectations, and use $\mathbb{E}[\widehat{g}\widehat{g}^T] = gg^T + \Sigma$. □

Thus changing a block LR preconditions the update; it does not by itself bias the underlying gradient estimator. For that block, the expected first-order movement scales as η_v, while the covariance of the SGD update scales as η_v^2. Smaller steps can suppress noisy motion but also delay progress. This is a local SGD statement. Adam's moment estimates are random and history-dependent, so it is not claimed as an Adam convergence theorem.

For Adam, the learning rate multiplies the normalized direction $\widehat{m}/(\sqrt{\widehat{s}} + \varepsilon)$ [2]. At fixed parameters, historical moments, and a fresh stochastic gradient, scaling that block's LR scales its parameter update. Multiplying its gradient by a constant is not equivalent: the first and second moments approximately cancel the scale when ε is negligible. We therefore implement actual parameter groups, not gradient multipliers. The legacy combined output layer is split into two parameter blocks and concatenated for the same forward Linear operation.

3.3 Separating backbones does not imply half the convergence speed

Both networks receive gradients from the same loss and both are updated at every iteration. There is no factor $1/2$ in either block's gradient. A simple counterexample rules out a necessary twofold slowdown: for $L(m, a) = (m^2 + a^2)/2$, joint SGD and two simultaneously updated separate blocks both obey $(m_{t+1}, a_{t+1}) =$

$(1-\eta)(m_t, a_t)$ exactly. A separate network can require more computation and can change conditioning and capacity; those facts do not imply a fixed change in the number of optimizer steps.

4 Why an arbitrarily slow variance network is not a complete solution

4.1 The initial scale must be able to shrink

The X-shaped target is

$$p = \tfrac{1}{2}N(0, \Sigma_+) + \tfrac{1}{2}N(0, \Sigma_-), \qquad \Sigma_\pm = \begin{pmatrix} 2 & \pm 1.8 \\ \pm 1.8 & 2 \end{pmatrix}.$$

Its component eigenvalues are 3.8 and 0.2. If S is an equiprobable sign, $T \sim N(0,1)$, and $U \sim N(0, I_2)$ independently, then

$$X = \sqrt{1.8}\, T(1, S)^T + \sqrt{0.2}\, U$$

has exactly this distribution. A global component variance of 0.2 is therefore sufficient with an unrestricted mixing distribution of means; a finite continuous neural network approximates the discrete branch selection.

Proposition 4.1 (An obstruction for frozen large component variance)**.** *No mixing distribution of means convolved with $N(0, vI_2)$ can equal this target when $v > 0.2$.*

Proof. In direction $w = (1,-1)^T/\sqrt{2}$, the target characteristic function is $\varphi_p(tw) = \frac{1}{2}e^{-0.1t^2} + \frac{1}{2}e^{-1.9t^2}$. A representation $X = M + \sqrt{v}U$ would require $\varphi_M(tw) = \varphi_p(tw)e^{vt^2/2}$, which is unbounded as $|t| \to \infty$ if $v > 0.2$. Every characteristic function has modulus at most one, a contradiction. □

More quantitatively, any such frozen-variance model satisfies

$$W_2(w^T q, w^T p) \geq W_1(w^T q, w^T p) \geq \sup_{t>0} \frac{[\varphi_p(tw) - e^{-vt^2/2}]_+}{t}.$$

The last inequality follows from the Lipschitz constant t of $x \mapsto e^{itx}$ and $|\varphi_q(tw)| \leq e^{-vt^2/2}$. For the initial variance 1.31326, our numerical one-dimensional bound is about 0.13535. It is a bound on this projection, not the averaged sliced metric, and does not explain the full magnitude of an escaping training run.

This distinguishes a hard floor of 0.2 from freezing the initial variance near 1.3. The floor permits shrinkage to the target's transverse scale; freezing the initial value prevents that adaptation. Initialization at 0.2 and its frozen control test this explanation directly.

4.2 A finite-budget obstruction for a very slow linear variance head

There is also a deterministic Adam bound that can make under-adaptation precise. Consider Adam without weight decay, initialized with zero moments, with $\beta_1^2 < \beta_2$ and nonnegative numerical ε. For any coordinate,

$$\left| \frac{\widehat{m}_t}{\sqrt{\widehat{s}_t} + \varepsilon} \right| \leq C := \frac{1-\beta_1}{\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}}. \tag{6}$$

For our betas, $C = 7.2703$. This bounds the coordinate update by $C\eta_t$ for every finite gradient sequence; no stochastic independence is assumed.

Proposition 4.2 (Reachability bound for a linear variance model)**.** *Let $a_j(Z) = w_j^T Z + b_j$, initially $w_j = 0$ and $b_j = 1$, with every variance parameter assigned LR $r\eta_t$. Put $\delta = Cr\sum_{t<T}\eta_t$. On the latent region $B_M = \{\|Z\|_1 \le M\}$, each variance is at least $\ell = \mathrm{softplus}(1-(1+M)\delta)$. Since $P(B_M) = \mathrm{erf}(M/2)^2 =: P_M$, regardless of the trained mean,*

$$W_2(w^T q_T, w^T p) \ge \sup_{t>0} \frac{[\varphi_p(tw) - P_M e^{-\ell t^2/2} - (1-P_M)]_+}{t}, \tag{7}$$

where $w = (1,-1)^T/\sqrt{2}$. A positive right-hand side rules out an exact fit within the specified optimization budget.

Proof. For completeness, Cauchy–Schwarz applied to the two weighted gradient moments gives the squared bound

$$C_t^2 = \frac{(1-\beta_1)^2}{(1-\beta_1^t)^2}\frac{1-\beta_2^t}{1-\beta_2}\frac{1-(\beta_1^2/\beta_2)^t}{1-\beta_1^2/\beta_2}.$$

The inequality $(1-x)(1-y) \le (1-\sqrt{xy})^2$, with $x = \beta_2^t$ and $y = (\beta_1^2/\beta_2)^t$, shows $C_t \le C$. Summing coordinate updates gives $|w_{jk}| \le \delta$ and $|b_j - 1| \le \delta$, so $a_j(Z) \ge 1-(1+M)\delta$ on B_M. Orthogonal rotation of a standard bivariate Gaussian gives $P(B_M) = \mathrm{erf}(M/2)^2$. Conditional noise in direction w has variance $(v_1+v_2)/2 \ge \ell$ on B_M. Therefore $|\varphi_{q_T}(tw)| \le P_M e^{-\ell t^2/2} + (1-P_M)$, independently of the conditional means. The Lipschitz characteristic-function argument from the previous subsection gives (7). □

For $r = 0.001$, $T = 50K$, and $M = 3$, the coordinate displacement is at most about 0.0723 and $P_M \simeq 0.9334$. The resulting projection bound is strictly positive. Retaining the original decay through 200K scarcely changes it. The recorded numerical checks give the full bounds for the restart policy and additional rates. These are bounds for the linear variance architecture, in exact arithmetic; they are not bounds for an unrestricted changing shared feature map. Linear-head runs at ratios 0.001 and 0.003 are included to test this predicted finite-budget limitation. The result shows why a universally beneficial prescription to decrease the variance LR cannot hold, even before considering stochastic gradient variance.

4.3 Representation noise can grow without a worse marginal

The conditional score has the exact coordinate second moment $\mathbb{E}[s_{c,j}^2] = \mathbb{E}[1/v_j(Z)]$. This is a diagnostic of representation noise, not a bound on every parameter-gradient variance or a proof of poor marginal fit. The following solvable example makes that distinction explicit.

Proposition 4.3 (An exact finite-batch gradient-noise example)**.** *Let $p = N(0,1)$ and, for $0 < v < 1$, represent $q_v = p$ as $X = \sqrt{1-v}\,E + \sqrt{v}\,U$, where E, U are independent standard Normals. Take the constant kernel $k = 1$, two independent batches of size N, and differentiate with respect to $r = \log v$ while preserving this representation. Both population objectives are zero for every v, but their stochastic gradient variances are*

$$\mathrm{Var}(\widehat{g}_C) = \frac{2+v-v^2}{2N^2v^2}, \tag{8}$$

$$\mathrm{Var}(\widehat{g}_S) = \frac{v}{2(1-v)N^2}. \tag{9}$$

Proof. Put $F = -X + U/\sqrt{v}$ and $B = \partial_r F$. Direct differentiation gives

$$F = -\sqrt{1-v}E + \frac{1-v}{\sqrt{v}}U, \quad B = \frac{v}{2\sqrt{1-v}}E - \frac{1+v}{2\sqrt{v}}U.$$

Hence $A = \mathrm{Var}(F) = (1-v)/v$, $C = \mathrm{Var}(B) = (1+v-v^2)/[4v(1-v)]$, and $D = \mathrm{Cov}(F,B) = -1/(2v)$. The original gradient is $\bar{B}\bar{F}' + \bar{F}\bar{B}'$, with independent primed batches, so its variance is $2(AC + D^2)/N^2$, giving the first expression. For the Stein form, $k = 1$ removes all spatial kernel derivatives and the loss is $\bar{X}\bar{X}'$. Here $\mathrm{Var}(X) = 1$, $\mathrm{Var}(\partial_r X) = v/[4(1-v)]$, and $\mathrm{Cov}(X, \partial_r X) = 0$. The same calculation gives the second expression. □

The constant kernel is an analytically solvable proxy, not the characteristic Gaussian kernel used in training. Independent fixed-bandwidth RBF simulations with the same unchanged marginal check whether the mechanism also appears with the training kernel family. The formulas prove a possible noise mechanism, not a universal RBF variance ordering: near $v = 0.8$ even the constant-kernel ordering reverses. A small learning rate changes update noise, while Stein integration removes the explicit component-score terms themselves.

Proposition 4.4 (Small-variance asymptotics for the Gaussian kernel)**.** *Let $p = q_v = N(0, I_d)$ be represented as $X_v = \sqrt{1-v}\,E + \sqrt{v}\,U$ with independent standard Gaussian vectors, and use two independent batches of fixed size N. For fixed $h > 0$ and $r = \log v$, the original Gaussian-kernel gradient satisfies*

$$\lim_{v \downarrow 0} v^2 \,\mathrm{Var}(\partial_r \widehat{L}_c) = \frac{d}{N^2}\left(1 + \frac{4}{h^2}\right)^{-d/2} > 0, \qquad \mathrm{Var}(\partial_r \widehat{L}_S) = O(v). \tag{10}$$

Both population gradients are zero for every v. Thus the noise divergence also occurs with a characteristic Gaussian kernel, without changing the marginal distribution or the population objective.

Proof. For the original estimator $F_v = -X_v + U/\sqrt{v}$, $F_v = U/\sqrt{v} + O_{L^p}(1)$ and $\partial_r F_v = -U/(2\sqrt{v}) + O_{L^p}(\sqrt{v})$ for every finite p. Also $\partial_r X_v = O_{L^p}(\sqrt{v})$. Gaussian kernel derivatives are bounded, so differentiation of its factor contributes only $O_{L^2}(v^{-1/2})$ to the gradient. Therefore

$$v\partial_r \widehat{L}_c \;\longrightarrow\; -\frac{1}{N^2}\sum_{i,j} k_h(E_i, E'_j) U_i^T U'_j \quad \text{in } L^2.$$

All cross terms except identical (i, j) vanish after conditioning on the latent E's and taking expectations over independent centered U's. The limiting variance is $d\mathbb{E}[k_h(E, E')^2]/N^2$. Since $E - E' \sim N(0, 2I_d)$, this kernel moment equals $(1 + 4/h^2)^{-d/2}$. The Stein integrand and its first spatial derivatives have uniformly bounded Gaussian moments, and its only r dependence is through X_v, Y_v. Its gradient is consequently $O_{L^2}(\sqrt{v})$, which gives the stated variance order. The L^2 limit also justifies passage of the mean and variance; the common population loss is identically zero. □

For $d = 2$, $N = 128$, and $h = 0.75$, the leading coefficient is approximately 1.5050×10^{-5}. Numerical simulations compare v^2 times the original gradient variance with this coefficient, rather than relying only on a visual log–log slope. This theorem concerns a specified marginal-preserving direction; it is not a lower bound for every parameter direction in a trained neural model.

5 Experimental protocol and atomic controls

All primary results use 50,000 updates. The fixed policy is Adam $(0.9, 0.999)$, mean LR 10^{-3}, two batches of 128, StepLR multiplier 0.9 per 1,000 updates, and score annealing $\alpha_t = 0.1 + 0.9 \min(t/25000, 1)$. The Gaussian distance arithmetic remains the StatComp implementation. Diagnostic RNG is isolated. We omit a diagnostic-only bandwidth-to-CPU synchronization in the fast update; equality of the loss, gradients, and production optimizer update is checked on CPU and GPU. No gradient clipping, EMA, or log-density regularization is introduced.

Discovery sweeps variance/mean LR ratios $\{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 3\}$ with seeds 42–44, for shared and separate width-128 two-layer backbones. For the shared model, only the variance output parameters receive this rate; for a separate model, the entire variance network receives it. Both networks are updated every step. Separate width-32, one-hidden-layer, linear, and shared stop-gradient alternatives are each tested at ratios $\{0.01, 0.1, 1\}$. Initial mean functions, constant variances, and training-noise seeds are paired.

The minimum mean discovery SW_2 at a positive LR selects one shared, one separate, and one alternative configuration. That selection is recorded before held-out seeds 45–49. Held-out controls include equal-LR shared and separate models, global variance, Stein loss, and a 0.2 floor. Further paired ablations change only annealing, initial variance, bandwidth detachment, or estimator/floor for the selected head LR. Every such change is labeled.

Training diagnostics use 10,000 samples and 128 projection directions. Independent final evaluation uses 20,000 samples and 128 directions, plus a second 10,000-sample check. Forward KL uses 2,048 exact target points and an empirical Gaussian mixture for q: mixture sizes 8,192, 16,384, and 65,536 are checked, and 262,144 is used when the last change exceeds 0.02. The last change is retained; the pointwise Monte Carlo standard error does not include all mixture-approximation uncertainty. Held-out KSD uses 16 independent cross-batch estimates, 256 samples per batch, at independently fixed bandwidths 0.5, 0.75, 1.5. Negative finite-sample estimates are retained.

5.1 Why update counts and learning-rate budgets are both reported

With $\eta_t = 10^{-3} 0.9^{\lfloor t/1000 \rfloor}$, the first 50,000 updates have total nominal LR

$$S_{50K} = 1000 \cdot 10^{-3} \frac{1 - 0.9^{50}}{1 - 0.9} = 9.94846.$$

The sum through 200,000 is almost 10, adding only 0.518% to S_{50K}. These sums are not actual Adam path lengths, but they show why extra decayed updates need not compensate for a smaller variance LR. We fork exact 50K checkpoints, including Adam and RNG state, to two labeled continuations: retained StepLR decay, or a restart at mean LR 10^{-4} with the original variance/mean LR ratio, held constant. The latter adds nominal mean LR 15 over the next 150K updates. Its improvement cannot be attributed to update count alone. All continuations are compared at 100K and 200K, and no run exceeds 200K updates.

6 Results and ablations

6.1 Learning-rate sweep: improvement has a finite useful range

On discovery seeds, the shared equal-LR mean error is 2.016, versus 0.178 at ratio 0.01. Shared ratios 0.003, 0.01, and 0.03 yield means 0.204, 0.178, and 0.185. This is a useful region rather than evidence that 0.01 is an exact optimum. Ratio 0.001 is more variable, and ratio zero fails. The independent width-128 two-layer variance network instead has mean errors 8.980, 4.239, and 0.122 at ratios 0.001, 0.003, and 0.03. High ratios can also fail; the separate ratio-3 mean is 4.562. Thus neither a monotone LR rule nor separation alone describes the observations.

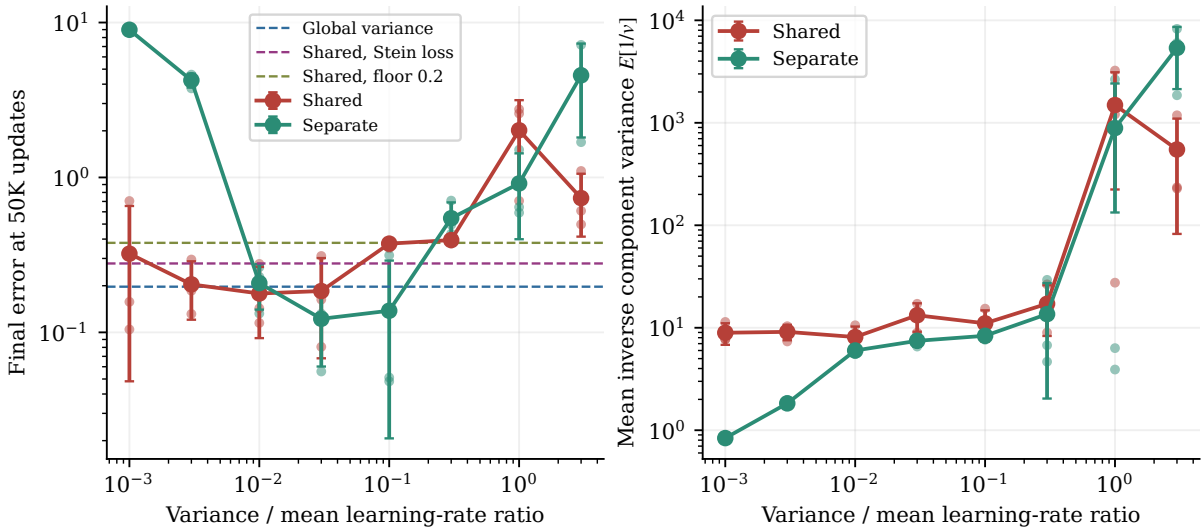


Figure 1. Three paired discovery seeds at 50K updates. Lines are arithmetic means; dots are runs. Error bars are sample SD, with lower ends clipped for logarithmic display only. The right panel measures component-score second moments through $\mathbb{E}[1/v]$, not marginal error. Zero-LR controls are tabulated in the appendix.

6.2 Prospective held-out comparison

Selection was committed at `808b97f` before seeds 45–49 were launched. The shared slow head improves all five paired equal-LR controls. It beats global variance on 3/5 seeds and Stein loss on 3/5. The separate selected network improves its own equal-LR control on 4/5 seeds, but has held-out mean 0.240, compared with 0.129 for the slow shared head. The selected linear variance map has mean 0.129, with all five errors between 0.101 and 0.145. These observations favor a slow shared output layer or restricted variance capacity as practical candidates. They do not show either candidate is statistically superior to global variance.

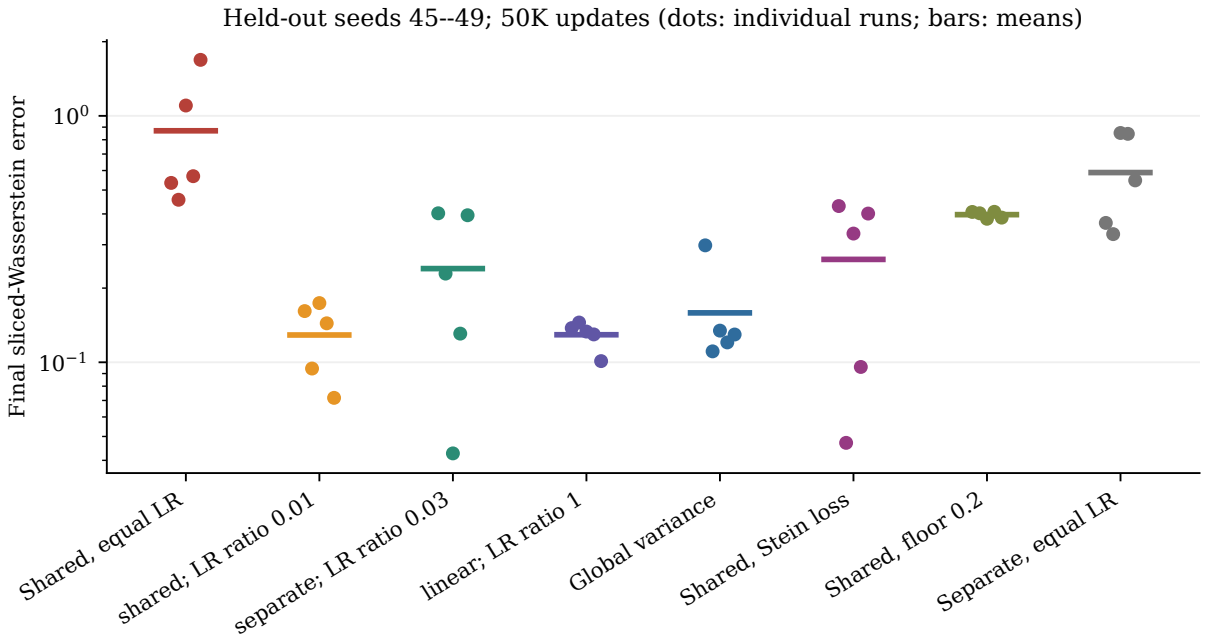


Figure 2. Five held-out training seeds, all at 50K optimizer updates. Each dot is one trained model. Horizontal bars are means. Hyperparameter selection uses none of these seeds.

Table 1. Paired held-out differences in training-diagnostic SW_2; negative favors the slow shared head.

Slow shared minus comparator	Mean difference	SD of differences	Wins
Shared, equal LR	-0.741	0.543	5/5
Separate 128x2; LR ratio 0.03	-0.111	0.185	3/5
Linear variance; LR ratio 1	-0.000	0.038	2/5
Global variance	-0.030	0.071	3/5
Shared, Stein loss	-0.132	0.175	3/5
Shared, floor 0.2	-0.268	0.044	5/5
Separate, equal LR	-0.460	0.235	5/5

6.3 Changing the variance backbone or its gradient route

The selected alternative is an independent affine map $a(Z) = WZ + c$ with only six variance parameters, versus two for global variance. It is a genuine conditional family, but the learned conditional variation is measured separately below. Its good result therefore need not imply that a rich conditional variance network was necessary. At ratio 0.01 the stop-gradient shared variant has mean error 6.682, compared with 0.178 with the full shared gradient. At ratio 0.1 it recovers to 0.365. Removing the variance-to-backbone gradient is therefore not a harmless way to implement a slower variance function: here it removes useful adaptation under the finite budget. The comparison is a causal routing intervention with identical forward architecture, but it also changes the update rule as described in Section 3.

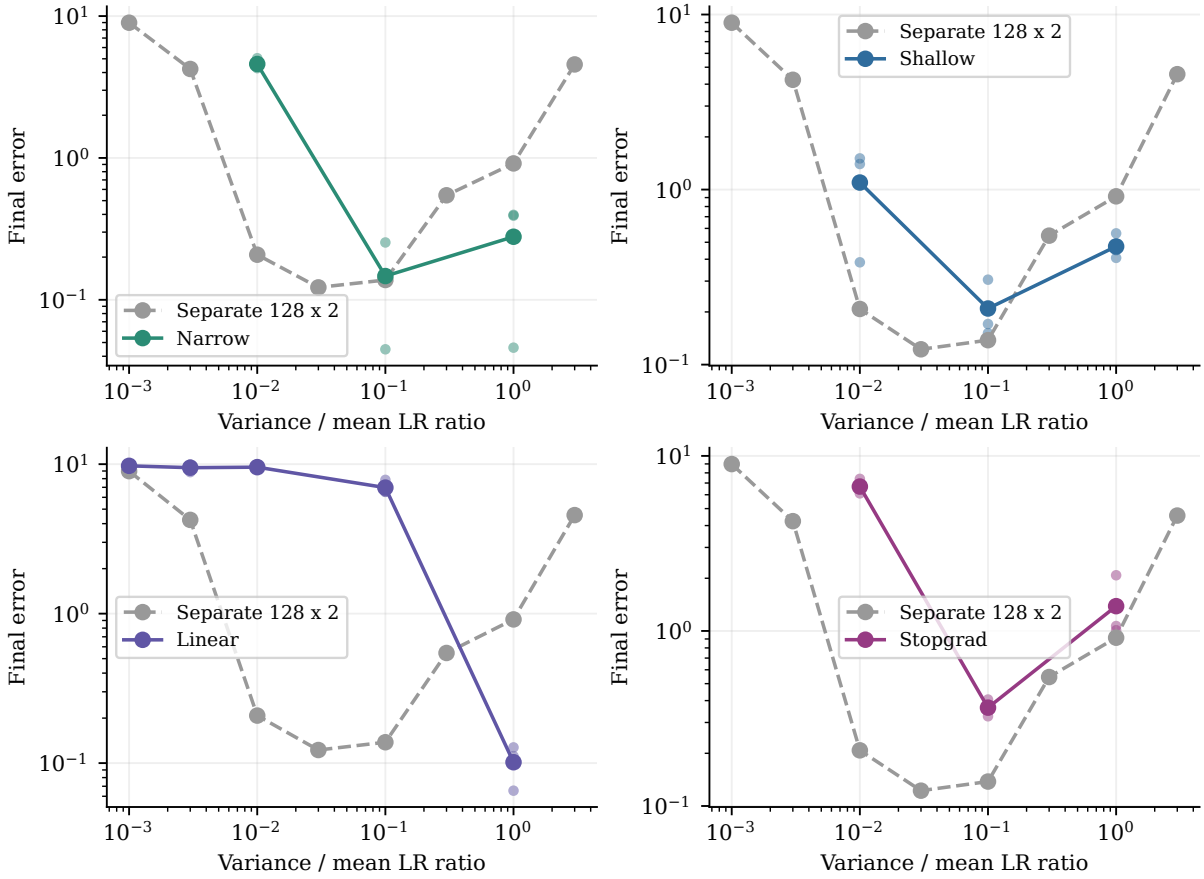


Figure 3. Variance architecture ablations, discovery seeds at 50K. The mean network remains width 128 with two hidden layers. Width 32, one hidden layer, and no hidden layer change only the variance branch relative to the separate network; stop-gradient changes only routing relative to shared features. Gray curves show the separate width-128 two-layer reference.

Table 2. Parameter counts. The common mean-only network has 17,154 parameters. For shared features, the last column counts additional variance output parameters; the backbone still receives both paths.

Variance architecture	Total parameters	Variance parameters
Global variance	17156	2
Shared, equal LR	17412	258
Separate, equal LR	34308	17154
Separate 32x2; LR ratio 1	18372	1218
Separate 128x1; LR ratio 1	17796	642
Linear variance; LR ratio 1	17160	6

6.4 Atomic initialization, annealing, kernel, and estimator controls

Initialization and freezing. Frozen variance 1.3133 gives mean error 9.467; frozen variance 0.2 gives 0.264. Initializing the selected separate network at 0.2 changes its mean from 0.122 to 0.064. This supports

the need to adapt the initially broad component noise. The effect is not uniform: initialization at 0.2 does not improve every selected shared or global run. A floor and a frozen initial scale are different interventions.

Annealing. Removing score annealing changes the shared equal-LR mean from 2.016 to 0.622, but changes global variance from 0.197 to 0.535, the selected separate network from 0.122 to 0.264, and Stein from 0.278 to 0.533. Annealing interacts with the optimizer state and model; disabling it is not a general cure.

Estimator and bandwidth. The slow-head original-loss mean is 0.178. Its bandwidth-detached control has mean 0.241, while Stein with that same slow head has mean 0.480. The latter two share the bandwidth-detachment policy, providing the cleaner estimator contrast. The tested slow head does not improve Stein. Adding a 0.2 floor to the slow original head gives mean 0.142, versus 0.378 for a floor at equal LR. This combination was evaluated only on discovery seeds; it is not a held-out recommendation.

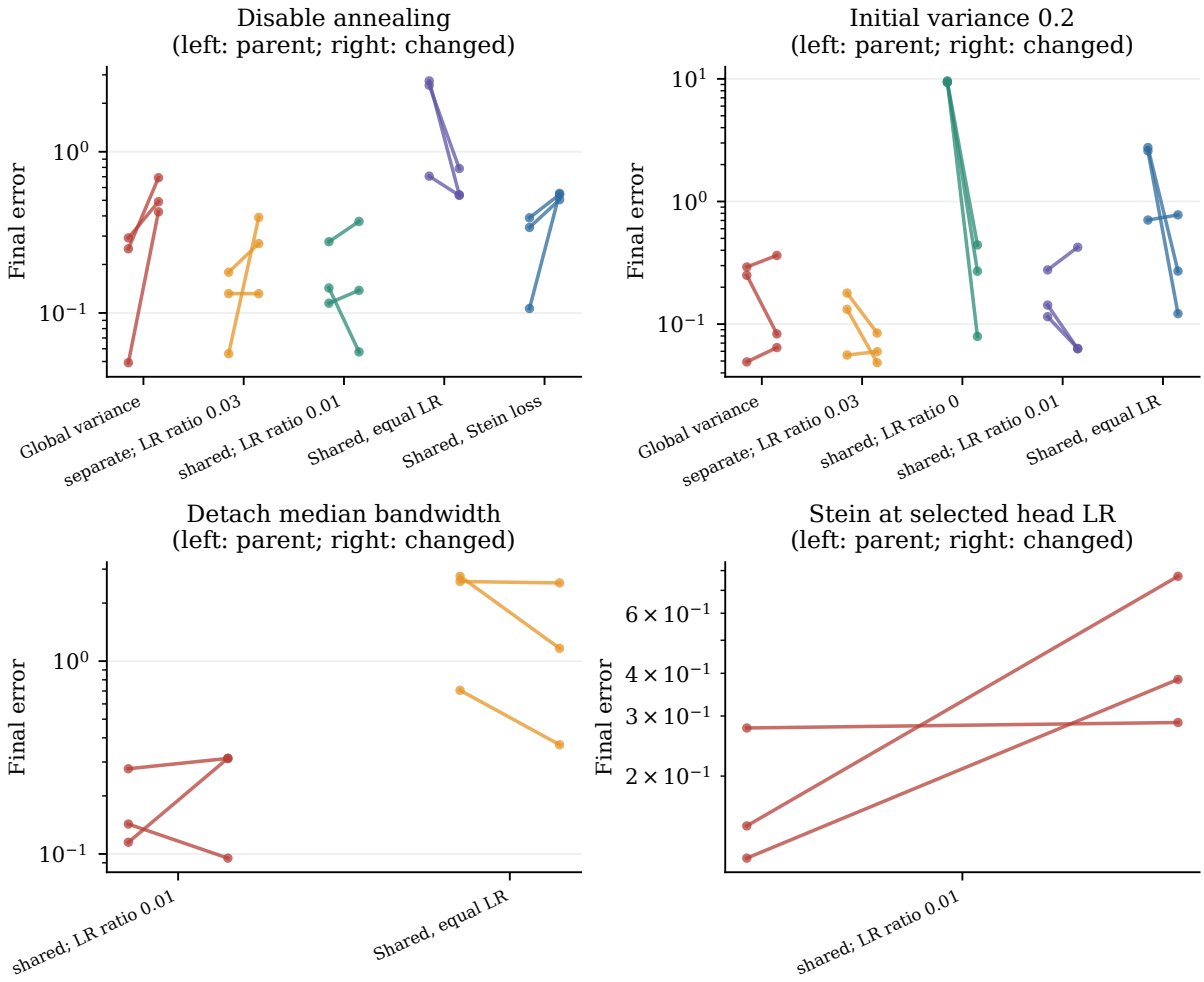


Figure 4. Paired atomic changes at 50K updates. Each short segment connects the same seed in the parent (left) and changed (right) configuration. A decrease is an improvement. The appendix includes all 14 contrasts, including the floor and frozen-variance controls.

6.5 50K, 100K, and 200K: continued decay versus restart

All continuations start from the exact 50K model, Adam moments, and training RNG state. Each branch adds 150K updates. The table and Figure 5 compare retained decay with a constant 10^{-4} mean LR restart; variance LR ratios remain fixed. Continuation evidence uses discovery seeds and should not be read as a new held-out ranking.

Table 3. Update-matched continuation results; means ± sample SD over three seeds. Restart changes cumulative LR as well as running longer.

Parent	Policy	50K	100K	200K
Global variance	decay	0.197 ± 0.130	0.192 ± 0.127	0.192 ± 0.127
Global variance	restart1e-4	0.197 ± 0.130	0.115 ± 0.025	0.103 ± 0.036
Linear variance; LR ratio 0.001	decay	9.747 ± 0.139	9.803 ± 0.135	9.803 ± 0.135
Linear variance; LR ratio 0.001	restart1e-4	9.747 ± 0.139	11.410 ± 0.567	17.826 ± 3.346
Separate 128x2; LR ratio 0.03	decay	0.122 ± 0.062	0.121 ± 0.059	0.121 ± 0.059
Separate 128x2; LR ratio 0.03	restart1e-4	0.122 ± 0.062	0.104 ± 0.044	0.109 ± 0.048
Shared; LR ratio 0.01	decay	0.178 ± 0.086	0.176 ± 0.086	0.176 ± 0.086
Shared; LR ratio 0.01	restart1e-4	0.178 ± 0.086	0.106 ± 0.003	0.109 ± 0.015
Shared, equal LR	decay	2.016 ± 1.138	1.985 ± 1.118	1.985 ± 1.118
Shared, equal LR	restart1e-4	2.016 ± 1.138	1.680 ± 0.755	1.461 ± 0.745
Shared, Stein loss	decay	0.278 ± 0.151	0.276 ± 0.154	0.276 ± 0.154
Shared, Stein loss	restart1e-4	0.278 ± 0.151	0.253 ± 0.176	0.225 ± 0.148

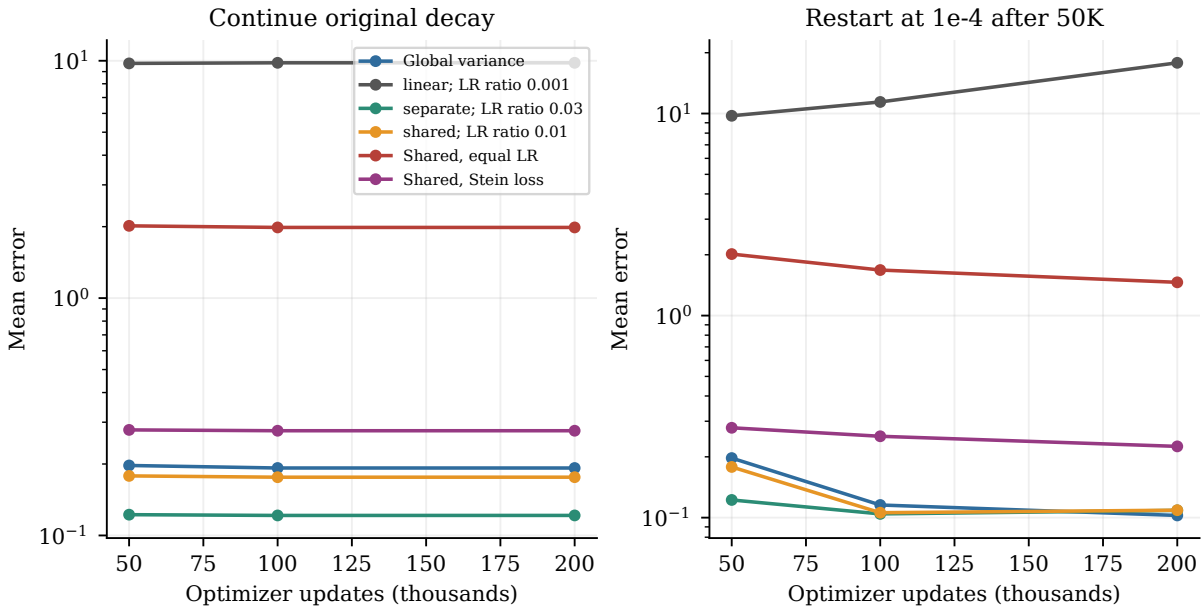


Figure 5. Matched checkpoint continuations. The original decay has almost exhausted its nominal LR budget at 50K. Restart supplies a materially different budget, while keeping the estimator and variance LR ratio unchanged. The linear ratio-0.001 control tests the finite-budget obstruction.

At 200K with restart, mean errors are 0.109 for the selected shared head, 0.109 for the selected separate network, 0.103 for global variance, and 0.225 for Stein. The shared equal-LR parent remains much worse at 1.461. The very slow linear ratio-0.001 model changes from 9.747 at 50K to 17.826; more optimization does not repair its frozen-scale-like constraint. The reachability theorem rules out an exact fit in this control but does not predict the large magnitude of its observed escape.

7 Mechanistic evidence and boundaries

7.1 Marginal fit and inverse-variance moments are distinct

The shared equal-LR discovery mean $\mathbb{E}[1/v]$ is 1490.1, compared with 8.12 for the slow head. Stein training nevertheless reaches 1488.0, with much better marginal error than the equal-LR original estimator. Conversely, the frozen wide-variance model has small inverse variance and poor fit. These controls reject an explanation based solely on the size of component variance: the score representation, update dynamics, and necessary variance adaptation all matter.

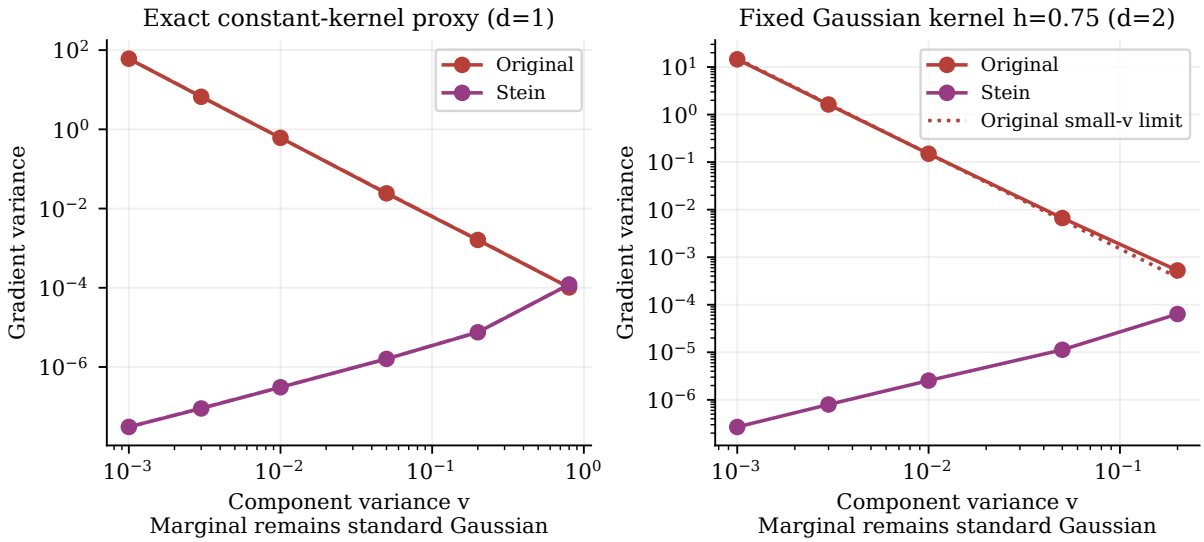


Figure 6. Unchanged-marginal controls: all representations have the same standard Gaussian marginal. Left: 20,000 replicated gradients and the exact constant-kernel formulas (dashed), two batches of 128. Right: 512 replications with fixed Gaussian bandwidth 0.75. The derivative follows log v while preserving the marginal; these are mechanism controls, not additional trained X-shaped models.

Increasing batch size can reduce this noise, but cannot give a variance-uniform guarantee at a fixed batch size. In the exact proxy, original variance behaves as $(Nv)^{-2}$ when v is small; keeping it bounded requires N to grow at least on the order of $1/v$. The Stein proxy instead has variance of order v/N^2 in that limit. This is an exact counterexample to a uniform fixed-batch cure, not a claim that every neural-network gradient has the same batch-size law. No new batch-size sweep is used to infer an X-shaped threshold.

The Gaussian-kernel theorem predicts $v^2 \operatorname{Var}(\widehat{g}_c) \to 1.50498 \times 10^{-5}$. The observed ratios to that limit are 0.9744 at $v = 0.003$, 0.9659 at $v = 0.001$. Thus the fixed-RBF simulation checks a proved coefficient, with the same Gaussian marginal throughout. A slower LR acts on the resulting update; it does not remove this estimator-level divergence.

7.2 Independent marginal metrics and density-approximation checks

Table 4. Independent evaluation of the same five held-out trained models. SD is across training seeds; KSD values are means of unbiased cross-batch estimates.

Held-out configuration	SW_2 (20K)	Forward KL	KSD^2, $h = .75$
Shared, equal LR	0.859 ± 0.496	0.895 ± 0.260	0.0162
Shared; LR ratio 0.01	0.127 ± 0.039	0.056 ± 0.049	-0.000461
Separate 128x2; LR ratio 0.03	0.236 ± 0.163	0.350 ± 0.388	0.000541
Linear variance; LR ratio 1	0.127 ± 0.013	0.061 ± 0.023	0.000554
Global variance	0.159 ± 0.079	0.121 ± 0.144	-5.09e-05
Shared, Stein loss	0.263 ± 0.175	0.296 ± 0.278	0.00314
Shared, floor 0.2	0.395 ± 0.011	0.531 ± 0.033	0.0049
Separate, equal LR	0.576 ± 0.219	0.661 ± 0.486	0.0102

The independent samples preserve the main comparisons without recycling the selection metric sample. Across all 262 evaluated checkpoints, 15 retain a last KL mixture-refinement change above 0.02; the maximum is 1.36. Those density estimates are flagged in the CSV and must not be treated as exact KL. None of the held-out 50K comparisons has a final refinement change above 0.02. Independent fixed-bandwidth KSD values and their Monte Carlo standard errors are retained at all three widths, including negative estimates. KSD, forward KL, and SW_2 measure different aspects of fit and need not order all bad runs identically. For example, if $q_R = (1-\epsilon)p + \epsilon r_R$, then $D_{KL}(p\|q_R) \leq -\log(1-\epsilon)$ even when r_R moves arbitrarily far away. Transport error can then grow without bound. This elementary example explains why a modest forward KL alone cannot rule out escaped mass; it does not assert that a particular checkpoint has this exact mixture form.

The linear variance model has held-out mean conditional coefficient of variation 0.0332 (averaged over dimensions and seeds; range 0.0143–0.0608). These small coefficients mean the learned variance is close to global on typical latent inputs. A good score for this low-capacity map is therefore not evidence that rich conditional variance was needed.

7.3 Gradient paths and functional drift at trained checkpoints

Table 5. Discovery checkpoints at 50K. Norms/cosines use the original loss and average 32 stochastic gradients per model and then training seeds. Freeze drift uses a standardized local mean LR 0.001, with the variance block frozen; it is an intervention, not an additional training run.

Checkpoint	Mean path norm	Variance path norm	Mean cosine	Freeze drift RMS
Shared, equal LR	14.4	5.99	0.013	0.0199
Shared; LR ratio 0.01	0.136	0.0206	-0.002	0.00232
Separate 128x2; LR ratio 0.03	0.103	0	0.000	0
Shared stop-gradient; LR ratio 0.01	0.116	0	0.000	0.0023
Linear variance; LR ratio 1	0.0797	0	0.000	0

The measured gradient decomposition has maximum reported mean relative residual 4.48e-07; the raw-output drift identity has maximum absolute residual 5.75e-14. A trained shared variance head can keep changing its output when its own LR is zero. The independent variance network has exactly zero drift when its entire block is frozen. Stop-gradient can still drift through mean-driven features. These interventions directly establish the remaining route. Gradient-path cosine alone would not prove it, and its sign is not a universal explanation of the training outcome.

Table 6. Unfrozen standardized local step and stochastic backbone gradients at the same checkpoints. Feature drift denotes the $W\Delta h$ term, including independent variance features for the separate network. Noise traces concern the original loss and only the mean backbone; a zero variance-path trace in separate/routed models is structural, not zero noise in their variance branch.

Checkpoint	Head drift RMS	Feature drift RMS	Mean-path noise trace	Var-path noise trace
Shared, equal LR	0.021	0.0199	467	75.8
Shared; LR ratio 0.01	0.000381	0.00232	0.0195	0.000544
Separate 128x2; LR ratio 0.03	0.000544	0.000613	0.0119	0
Shared stop-gradient; LR ratio 0.01	0.000249	0.0023	0.0158	0
Linear variance; LR ratio 1	0.000318	0	0.00722	0

The slow shared head has feature-driven raw-variance drift larger than head-driven drift in this local intervention, confirming that its variance function is not simply frozen. Equal-LR shared checkpoints have much larger measured backbone noise traces than the selected slow-head checkpoints. These are comparisons between different learned states: they support the noise mechanism but do not isolate a same-state causal effect of the LR.

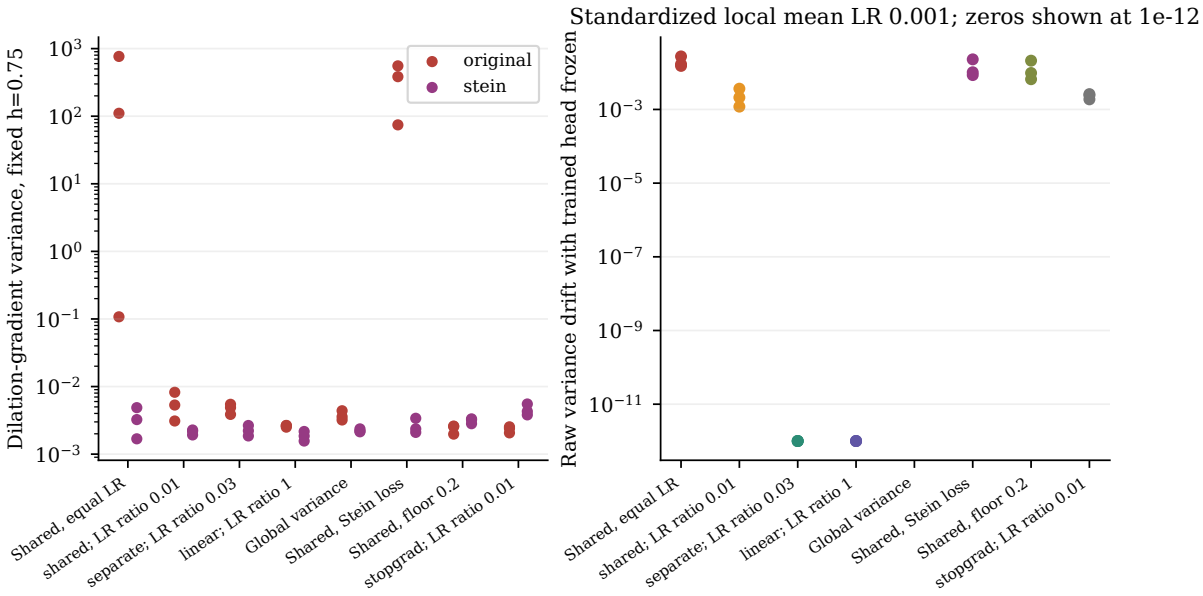


Figure 7. Trained-checkpoint probes on discovery seeds. Left: gradient variance for a common marginal-dilation direction, using independently fixed bandwidth 0.75 and 128 repeated pairs of batches. Right: raw variance output drift after freezing the trained variance block, with mean LR 0.001; exact zeros are displayed at 10^{-12}. Parameter-gradient and semantic-dilation diagnostics have different scopes.

8 Claim-to-evidence audit and practical interpretation

Claim	Evidence	Boundary
Slow shared head improves the tested original loss.	LR-only paired intervention; prospective five-seed confirmation.	This target, transform, mean architecture, batch size, and 50K policy.
Head LR cannot control all shared variance motion.	Exact identity (4), gradient decomposition, and trained-head freeze intervention.	A route exists; no universal nonzero lower bound at every state.
Very slow variance learning can be insufficient.	Frozen-scale characteristic-function obstruction; finite-budget Adam bound; linear tiny-LR and initialization controls.	The quantitative reachability theorem is for the linear variance architecture.
Small component variance can increase original gradient noise without worse marginal fit.	Exact constant-kernel formula, Gaussian-kernel asymptotic theorem, and unchanged-marginal replications.	Specified marginal-preserving direction; not a universal covariance ordering or every parameter-gradient variance.
Separate backbones do not necessarily halve convergence speed.	Exact simultaneous-quadratic counterexample; update counts matched in all comparisons.	Compute and conditioning can change; no universal wall-time speed guarantee.
Different block LRs do not themselves bias the estimator.	Block-preconditioning identity and local smoothness inequality.	SGD local statement; no Adam global convergence claim.
Slowing the head and using Stein are different interventions.	Original-detached versus Stein at the same slow LR; fixed-width noise probes.	Stein default comparison also changes bandwidth differentiation.
Extra decayed steps provide little nominal LR budget.	Geometric-series calculation and checkpoint continuation/restart controls.	Nominal LR sums are not actual Adam path lengths.

8.1 Practical interpretation

For the original conditional-variance loss and this training policy, start with a shared variance output LR 10^{-5} and mean/backbone LR 10^{-3}. This is an empirically supported candidate, not a proven universal optimum. Keep the backbone gradient intact; a smaller output LR already changes its learned trajectory. If using an independent variance branch, tune its rate again rather than transferring the shared ratio unchanged. A very small branch may be sufficient; the linear branch at ratio 1 is competitive here. Compare at matched optimizer steps and report whether LR was restarted. Use inverse-variance moments alongside independent marginal metrics; neither diagnostic alone identifies the whole mechanism.

8.2 Limitations

Discovery minima are selected estimates, so held-out results carry the main weight in recommendations. Five held-out seeds give limited precision for rare training failures. The investigation covers one two-dimensional target, one batch size, one mean architecture, one variance transform, and finite training budgets. Shared versus separate backbones change parameterization and capacity; narrow, shallow, and linear variants examine but do not eliminate all such differences. Gradient-path correlations and inverse-variance moments are state diagnostics, not alone proofs of causal responsibility. The freeze intervention, paired training ablations, and exact formulas provide different forms of evidence and are identified separately. Adaptive bandwidths and annealing limit direct application of fixed-population SGD theory; no global convergence theorem for the implemented Adam procedure is claimed.

A Complete numerical comparisons

A.1 Full shared/separate LR sweep

Table 8. Final training-diagnostic SW_2, 10K samples and 128 fixed projection directions, at 50K updates. All seeds are shown.

Configuration	42	43	44	Mean ± SD
Shared, floor 0.2	0.355	0.385	0.394	0.378 ± 0.020
Global variance	0.292	0.049	0.250	0.197 ± 0.130
Legacy conditional	2.754	2.588	0.706	2.016 ± 1.138
Separate 128x2; LR ratio 0	9.434	9.636	9.331	9.467 ± 0.155
Separate 128x2; LR ratio 0.001	9.367	8.767	8.806	8.980 ± 0.336
Separate 128x2; LR ratio 0.003	4.608	3.759	4.348	4.239 ± 0.435
Separate 128x2; LR ratio 0.01	0.263	0.228	0.132	0.208 ± 0.068
Separate 128x2; LR ratio 0.03	0.056	0.132	0.179	0.122 ± 0.062
Separate 128x2; LR ratio 0.1	0.048	0.051	0.314	0.138 ± 0.153
Separate 128x2; LR ratio 0.3	0.433	0.495	0.709	0.545 ± 0.145
Separate, equal LR	0.643	1.510	0.592	0.915 ± 0.516
Separate 128x2; LR ratio 3	7.174	4.823	1.690	4.562 ± 2.752
Shared; LR ratio 0	9.434	9.636	9.331	9.467 ± 0.155
Shared; LR ratio 0.001	0.104	0.157	0.705	0.322 ± 0.332
Shared; LR ratio 0.003	0.295	0.131	0.186	0.204 ± 0.083
Shared; LR ratio 0.01	0.115	0.143	0.277	0.178 ± 0.086
Shared; LR ratio 0.03	0.163	0.311	0.081	0.185 ± 0.117
Shared; LR ratio 0.1	0.391	0.382	0.349	0.374 ± 0.022
Shared; LR ratio 0.3	0.394	0.375	0.414	0.394 ± 0.020
Shared, equal LR	2.754	2.588	0.706	2.016 ± 1.138
Shared; LR ratio 3	0.609	1.101	0.498	0.736 ± 0.321
Shared, Stein loss	0.107	0.389	0.339	0.278 ± 0.151

A.2 Variance architecture and routing sweep

Table 9. Final training-diagnostic SW_2, 10K samples and 128 fixed projection directions, at 50K updates. All seeds are shown.

Configuration	42	43	44	Mean ± SD
Linear variance; LR ratio 0.001	9.830	9.586	9.824	9.747 ± 0.139
Linear variance; LR ratio 0.003	8.848	9.626	9.944	9.472 ± 0.564
Linear variance; LR ratio 0.01	9.277	10.013	9.354	9.548 ± 0.404
Linear variance; LR ratio 0.1	7.850	6.527	6.529	6.969 ± 0.763
Linear variance; LR ratio 1	0.111	0.127	0.065	0.101 ± 0.032
Separate 32x2; LR ratio 0.01	5.048	4.294	4.439	4.593 ± 0.400
Separate 32x2; LR ratio 0.1	0.045	0.143	0.254	0.147 ± 0.104
Separate 32x2; LR ratio 1	0.046	0.397	0.392	0.278 ± 0.201
Separate 128x1; LR ratio 0.01	0.384	1.507	1.400	1.097 ± 0.620
Separate 128x1; LR ratio 0.1	0.306	0.151	0.171	0.209 ± 0.084
Separate 128x1; LR ratio 1	0.407	0.450	0.562	0.473 ± 0.080
Shared stop-gradient; LR ratio 0.01	7.395	6.543	6.108	6.682 ± 0.655
Shared stop-gradient; LR ratio 0.1	0.406	0.325	0.363	0.365 ± 0.040
Shared stop-gradient; LR ratio 1	1.010	2.081	1.067	1.386 ± 0.603

A.3 Prospective held-out training seeds

Table 10. Final training-diagnostic SW_2, 10K samples and 128 fixed projection directions, at 50K updates. All seeds are shown.

Configuration	45	46	47	48	49	Mean ± SD
Shared, floor 0.2	0.407	0.402	0.382	0.408	0.386	0.397 ± 0.012
Global variance	0.298	0.111	0.134	0.120	0.130	0.159 ± 0.079
Linear variance; LR ratio 1	0.138	0.145	0.133	0.130	0.101	0.129 ± 0.017
Separate 128x2; LR ratio 0.03	0.402	0.229	0.043	0.131	0.395	0.240 ± 0.159
Separate, equal LR	0.368	0.331	0.852	0.845	0.547	0.589 ± 0.251
Shared; LR ratio 0.01	0.161	0.094	0.174	0.144	0.072	0.129 ± 0.044
Shared, equal LR	0.534	0.457	1.102	0.569	1.688	0.870 ± 0.524
Shared, Stein loss	0.431	0.047	0.333	0.096	0.401	0.262 ± 0.178

A.4 Every atomic ablation

Table 11. Final training-diagnostic SW_2, 10K samples and 128 fixed projection directions, at 50K updates. All seeds are shown.

Configuration	42	43	44	Mean ± SD
global init02	0.363	0.064	0.083	0.170 ± 0.167
global noanneal	0.490	0.424	0.690	0.535 ± 0.139
Separate 128x2; LR ratio 0.03 init02	0.060	0.048	0.085	0.064 ± 0.019
Separate 128x2; LR ratio 0.03 noanneal	0.392	0.132	0.270	0.264 ± 0.130
Shared; LR ratio 0 init02	0.442	0.079	0.270	0.264 ± 0.181
Shared; LR ratio 0.01 detach	0.313	0.095	0.313	0.241 ± 0.126
Shared; LR ratio 0.01 floor02	0.102	0.179	0.143	0.142 ± 0.039
Shared; LR ratio 0.01 init02	0.063	0.063	0.424	0.183 ± 0.208
Shared; LR ratio 0.01 noanneal	0.138	0.057	0.370	0.188 ± 0.162
Shared; LR ratio 0.01 stein	0.384	0.770	0.287	0.480 ± 0.255
Shared; LR ratio 1 detach	1.165	2.544	0.369	1.359 ± 1.101
Shared; LR ratio 1 init02	0.271	0.122	0.778	0.390 ± 0.344
Shared; LR ratio 1 noanneal	0.541	0.787	0.536	0.622 ± 0.143
stein noanneal	0.552	0.543	0.505	0.533 ± 0.025

A.5 All continuation endpoints

Table 12. Every 200K endpoint; the initial 50K updates are shared with the corresponding parent.

Parent and policy	42	43	44	Mean ± SD
Global variance; decay	0.286	0.048	0.242	0.192 ± 0.127
Global variance; restart1e-4	0.121	0.061	0.125	0.103 ± 0.036
Linear variance; LR ratio 0.001; decay	9.892	9.647	9.870	9.803 ± 0.135
Linear variance; LR ratio 0.001; restart1e-4	20.245	19.227	14.008	17.826 ± 3.346
Separate 128x2; LR ratio 0.03; decay	0.057	0.133	0.174	0.121 ± 0.059
Separate 128x2; LR ratio 0.03; restart1e-4	0.055	0.125	0.146	0.109 ± 0.048
Shared; LR ratio 0.01; decay	0.114	0.140	0.274	0.176 ± 0.086
Shared; LR ratio 0.01; restart1e-4	0.100	0.100	0.126	0.109 ± 0.015
Shared, equal LR; decay	2.703	2.555	0.696	1.985 ± 1.118
Shared, equal LR; restart1e-4	2.191	1.490	0.702	1.461 ± 0.745
Shared, Stein loss; decay	0.100	0.387	0.341	0.276 ± 0.154
Shared, Stein loss; restart1e-4	0.055	0.327	0.293	0.225 ± 0.148

A.6 Component variance diagnostics on discovery runs

Table 13. Arithmetic means of per-model diagnostics over three discovery seeds. Quantiles pool latent samples and coordinates within a model; they are not quantiles across training seeds.

Configuration	$\mathbb{E}[1/v]$	1% quantile	Median	Mean
Shared, floor 0.2	4.61	0.2	0.2	0.231
Global variance	5.11	0.195	0.195	0.196
Legacy conditional	1.49e+03	0.000891	0.183	0.331
Separate 128x2; LR ratio 0	0.761	1.31	1.31	1.31
Separate 128x2; LR ratio 0.001	0.838	1.13	1.2	1.19
Separate 128x2; LR ratio 0.003	1.83	0.292	0.591	0.567
Separate 128x2; LR ratio 0.01	6	0.0618	0.197	0.188
Separate 128x2; LR ratio 0.03	7.45	0.0452	0.168	0.16
Separate 128x2; LR ratio 0.1	8.35	0.0402	0.149	0.142
Separate 128x2; LR ratio 0.3	13.6	0.0284	0.235	0.248
Separate, equal LR	889	0.0403	0.215	0.296
Separate 128x2; LR ratio 3	5.38e+03	0.0001	0.0642	0.304
Shared; LR ratio 0	0.761	1.31	1.31	1.31
Shared; LR ratio 0.001	8.94	0.0196	0.191	0.187
Shared; LR ratio 0.003	9.12	0.019	0.183	0.184
Shared; LR ratio 0.01	8.12	0.0392	0.161	0.163
Shared; LR ratio 0.03	13.2	0.0117	0.153	0.154
Shared; LR ratio 0.1	11.1	0.0171	0.153	0.166
Shared; LR ratio 0.3	17.1	0.00991	0.171	0.18
Shared, equal LR	1.49e+03	0.000891	0.183	0.331
Shared; LR ratio 3	549	0.000103	0.156	0.362
Shared, Stein loss	1.49e+03	0.0001	0.0871	0.171
Linear variance; LR ratio 0.001	0.765	1.31	1.31	1.31
Linear variance; LR ratio 0.003	0.771	1.29	1.3	1.3
Linear variance; LR ratio 0.01	0.794	1.25	1.26	1.26
Linear variance; LR ratio 0.1	1.19	0.72	0.844	0.845
Linear variance; LR ratio 1	5.08	0.177	0.197	0.197
Separate 32x2; LR ratio 0.01	1.81	0.276	0.596	0.583
Separate 32x2; LR ratio 0.1	7.55	0.0344	0.171	0.163
Separate 32x2; LR ratio 1	7.39	0.0568	0.151	0.179
Separate 128x1; LR ratio 0.01	3.19	0.118	0.372	0.353
Separate 128x1; LR ratio 0.1	6.67	0.0457	0.183	0.174
Separate 128x1; LR ratio 1	22.2	0.00702	0.187	0.187
Shared stop-gradient; LR ratio 0.01	1.12	0.368	0.96	0.941
Shared stop-gradient; LR ratio 0.1	7.08	0.0437	0.212	0.234
Shared stop-gradient; LR ratio 1	310	0.0001	0.344	0.366

A.7 Finite-budget theoretical numerical checks

Table 14. Numerical maximization of the proved projection bound, $P(B_3) = 0.933359$. A zero bound is inconclusive, not a guarantee of good optimization.

Ratio and policy	Coordinate bound	ℓ on B_3	Projected W_2 bound
0.001; 50k	0.07233	1.11034	0.08375
0.001; 200k_decay	0.07270	1.10933	0.08365
0.001; 200k_restart1e-4	0.18138	0.83977	0.05512
0.003; 50k	0.21698	0.76136	0.04601
0.003; 200k_decay	0.21811	0.75896	0.04573
0.003; 200k_restart1e-4	0.54415	0.26875	0.00000
0.01; 50k	0.72328	0.14028	0.00000
0.01; 200k_decay	0.72703	0.13833	0.00000
0.01; 200k_restart1e-4	1.81383	0.00192	0.00000

B Final sample distributions

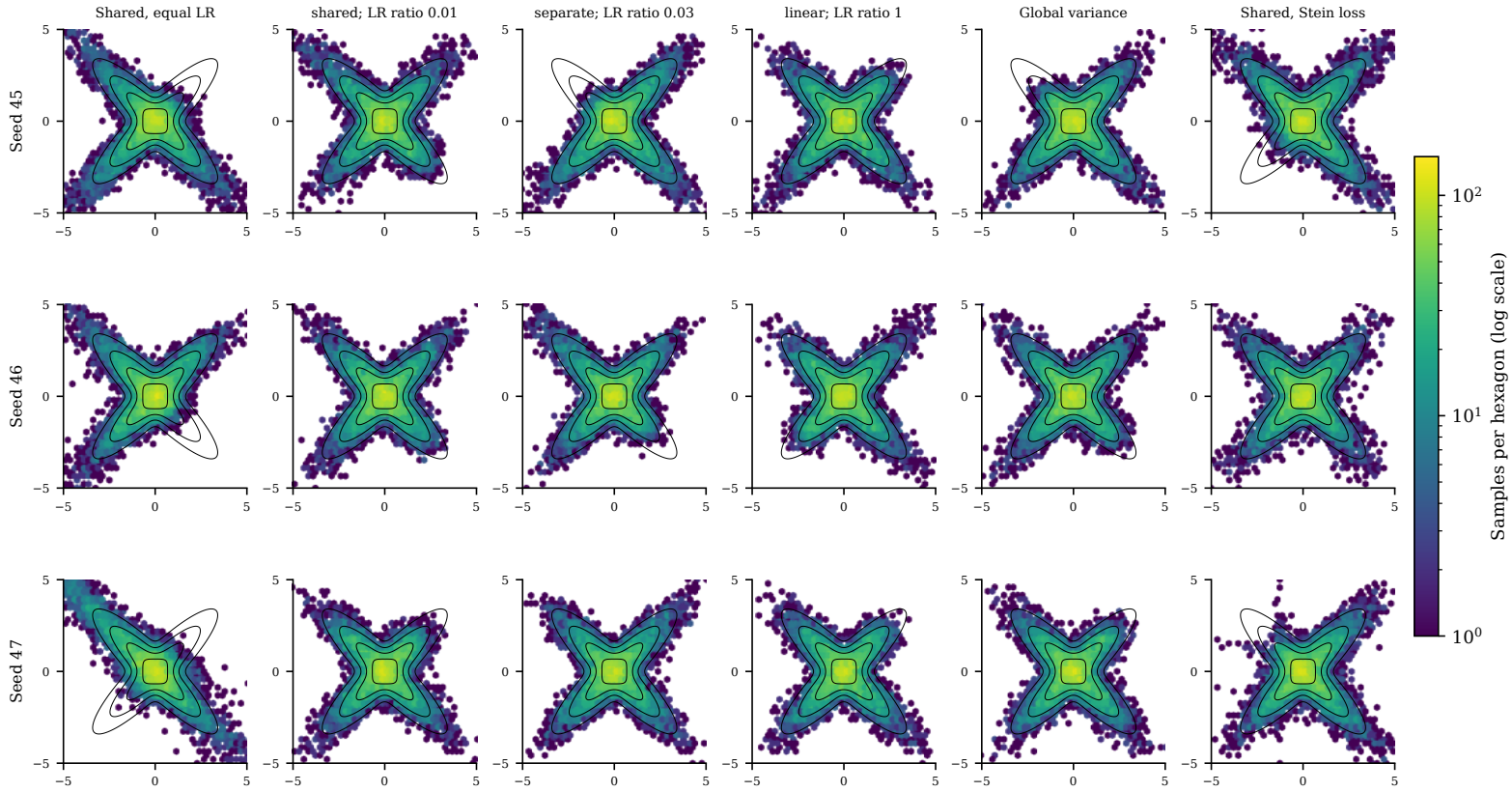


Figure 8. The first three held-out seeds (45–47), with configurations fixed from discovery. Each panel shows all 10K final samples within the common viewing window, with exact target contours in black and a common logarithmic count scale. Metrics include samples outside the window. Columns: equal-LR conditional, slow shared, selected separate, linear variance, global variance, Stein loss.

C Reproduction, validation, and artifact inventory

The final campaign contains 226 completed training trajectories: 66 sweep, 42 architecture, 40 held-out, 42 ablation, and 36 continuation branches. Counting shared parent updates once gives 14,900,000 optimizer updates. The 36 continuations each add 150K updates; no trajectory exceeds 200K. Experiments run on an RTX 4090 using PyTorch 2.9.0+cu126 in the server ruivi environment. Each source manifest records the commit, relevant SHA-256 hashes, complete model configuration, parameter count, and training specification.

Final audit. 262 endpoint/checkpoint sample arrays and diagnostics were regenerated. Loss gradients were compared against an independent kernel expression, with the Stein operator computed by spatial autograd; maximum relative difference 5.72e-07. The split-head equal-LR and legacy conditional models have bitwise identical parameters at all 18 checked seed/checkpoint pairs, and all 36 continuation branches preserve their parent model, RNG, and Adam moments at 50K. Production-runner update equality and parameter coverage are recorded separately in `validation.json`.

Saved artifacts. `training_results.json` and the metric/trajectory CSVs contain every run; `selection.json` records the discovery selection; `evaluation.json` and `independent_metrics.csv` contain all independent SW_2, KL refinement, and fixed-bandwidth KSD estimates and standard errors; `theory_checks.json` and `audit.json` retain numerical checks and endpoint verification. Figures are supplied as vector PDF and PNG. Raw model/optimizer/RNG checkpoints and samples remain under `/root/ruivi/results/ksivi_variance_lr_20261011`.

Production configuration. Use `ConditionalGaussianHeads` to enable real parameter groups. The legacy `ConditionalGaussian` retains its original combined optimizer behavior. The new shared model can be selected from the existing KSIVI config with these overrides (the first line is the model type, followed by optimizer settings):

```
vi_model_type=ConditionalGaussianHeads
train.vi.lr=0.001 train.vi.var_lr=0.00001
vi_model.variance_backbone=shared
```

For the selected separate network use `variance_backbone=separate` and `var_lr=0.00003`. For the linear variance branch set `variance_num_layers=0` and `var_lr=0.001`. Full commands, phase order, evaluation, report generation, and source changes are documented in the campaign README. Existing canonical configs are not overwritten.

References

[1] Z. Cheng, L. Yu, T. Xie, S. Zhang, and C. Zhang. *Kernel Semi-Implicit Variational Inference*. ICML, 2024. https://arxiv.org/abs/2405.18997.

[2] D. P. Kingma and J. Ba. *Adam: A Method for Stochastic Optimization*. ICLR, 2015. https://arxiv.org/abs/1412.6980.